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Annual Review of Animal Biosciences

Evolution of Vertebrate Hormones and Their Receptors: Insights from Non-Osteichthyan Genomes

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Annu. Rev. Anim. Biosci. 2023. 11:163–82

First published as a Review in Advance on November 18, 2022

The *Annual Review of Animal Biosciences* is online at animal.annualreviews.org

https://doi.org/10.1146/annurev-animal-050922-071351

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Keywords

leptin, oxytocin, vasopressin, prolactin, ghrelin, cartilaginous fishes, cyclostomes

Abstract

Homeostatic control and reproductive functions of humans are regulated at the molecular levels largely by peptide hormones secreted from endocrine and/or neuroendocrine cells in the central nervous system and peripheral organs. Homologs of those hormones and their receptors function similarly in many vertebrate species distantly related to humans, but the evolutionary history of the endocrine system involving those factors has been obscured by the scarcity of genome DNA sequence information of some taxa that potentially contain their orthologs. Focusing on non-osteichthyan vertebrates, namely jawless and cartilaginous fishes, this article illustrates how investigating genome sequence information assists our understanding of the diversification of vertebrate gene repertoires in four broad themes: (*a*) the presence or absence of genes, (*b*) multiplication and maintenance of paralogs, (*c*) differential fates of duplicated paralogs, and (*d*) the evolutionary timing of gene origins.

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1. INTRODUCTION

Homeostatic processes and reproduction of humans and other mammals are regulated by molecular signaling from endocrine and/or neuroendocrine cells to target tissues including central nervous systems and peripheral organs. This signaling is achieved through an intricate network of various hormones, namely, peptides/proteins or small-molecule compounds recognized by receptors expressed as membrane or cytoplasmic/nuclear proteins in cells in their target tissues. Since the recognition of the basic repertoires of those bioactive substances acting as hormones, their functions have been explored in diverse organisms, including nonmammalian vertebrates and invertebrate animals (1).

For hormones and their receptors, cross-species comparisons of amino acid sequences or nucleotide sequences of the genes encoding them in phylogenetic contexts allow us to infer the evolutionary processes of the individual endocrine system components. Morphological studies, often involving developmental biology approaches, provide profound clues for the origins of cross-species differences and commonalities of the organs containing endocrine cells but sometimes do not coherently address the question of homology. Fossils cannot preserve the physiology of extinct animals, but our understanding of the evolutionary processes of individual components of the molecular network can fill this gap. For example, the evolutionary point at which cross-talk between different organs, such as the gut–brain axis (2), arose remains unaddressed.

Evolutionary processes have long been inferred with an emphasis on molecular functions of individual hormones and receptors, which has consolidated our knowledge of the evolution of vertebrate endocrine systems. This knowledge is now being enriched through cataloging those components by comprehensive surveys of gene repertoires and the tissues in which they are expressed. Protein-coding genes of common origins (homologs) are categorized into those derived from speciation (orthologs) and those derived from gene duplication (paralogs), which have been investigated routinely through multiple sequence alignment and molecular phylogeny inference. One recent study of this kind identified shark orthologs of many mammalian genes encoding hormones and their receptors in whole-genome sequences (**Figure 1**), which suggested that the genetic machinery governing mammalian homeostasis and reproduction, such as the gut–brain axis, was established in the common ancestor of all extant jawed vertebrates (gnathostomes) (3).

This review provides general explanations of how molecular phylogenetics and genome informatics contribute to our understanding of the vertebrate endocrine system, as well as some examples for differentiating the gene repertoires of hormones and their receptors during vertebrate evolution. It provides an overview of the possible organization of the endocrine machinery in early vertebrates.

2. FISH ARRIVED LATE IN GENETIC RESEARCH BUT APPEARED EARLIER IN EVOLUTION

At the dawn of the twentieth century, Sharpey-Schafer invented the very concept of internal secretion or endocrinology; shortly thereafter, Starling named secretin as the first hormone (4). Their insights have opened a century of hormone discovery, first by bioassay and structure determination and then by advanced chromatographic purification and radioimmunoassay in large mammals (5). More recently, studies on humans or rodents have become more predominant with the rise of molecular genetic approaches. Our knowledge was expanded by further studies on nonmammalian model systems, including chicken and zebrafish, based on sequence homology of molecules. Such studies usually used animals commonly regarded as nonmodel species, such as sheep, pig, goldfish, and salmon, all of which are either mammals or actinopterygian fishes that are classified as osteichthyans (bony vertebrates).

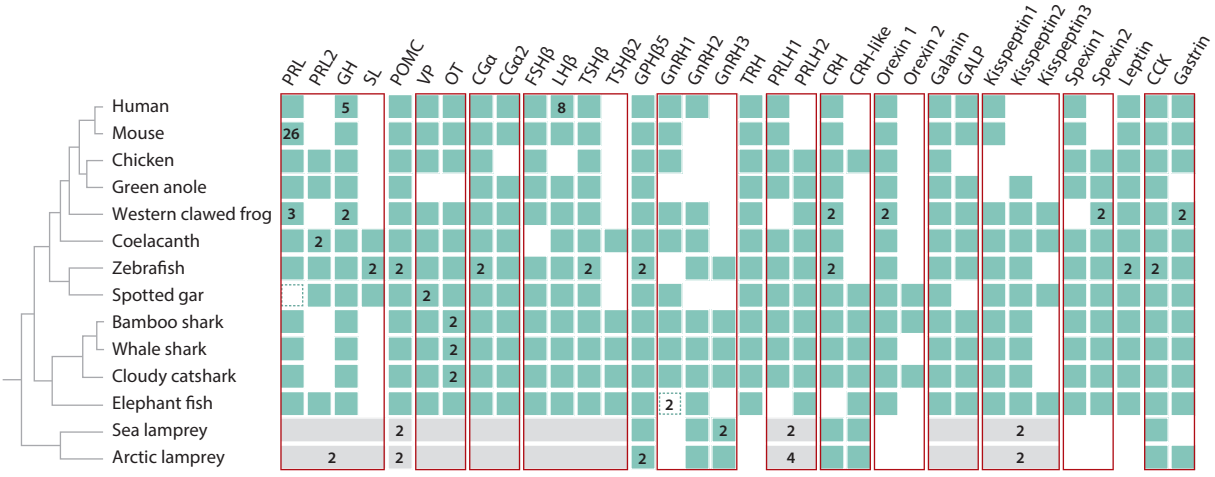

Figure 1

Cross-species comparison of some vertebrate hormone gene repertoires. The numbers of paralogs are indicated in the boxes based on a previous study (3). Represented are the genes whose orthology was inferred (*green boxes*), those with ambiguous orthology among multiple subtype genes (*gray boxes*), and members of the same gene families (*red rectangles*). Abbreviations: CCK, cholecystokinin; CG, chorionic gonadotropin; CRH, corticotropin-releasing hormone; FSH, follicle-stimulating hormone; GALP, galanin-like peptide; GH, growth hormone; GnRH, gonadotropin-releasing hormone; GPH, glycoprotein hormone; LH, luteinizing hormone; OT, oxytocin; POMC, pro-opiomelanocortin; PRL, prolactin; PRLH, prolactin-releasing hormone; SL, somatolactin; TRH, thyrotropin-releasing hormone; TSH, thyroid-stimulating hormone; VP, vasopressin.

Along with the rise of large-scale sequencing technologies, in silico sequence identification rapidly prevailed to cover species evolutionarily distant from mammals, as well as those with low tissue sample availability. Although collectively denoted as a paraphyletic group of fishes, extant non-tetrapod lineages of vertebrates are divided into several groups exhibiting distinct phenotypic traits (**Figure 2**). Among those fish lineages, jawless fishes (cyclostomes) and cartilaginous fishes (chondrichthyans) are recognized as key taxa in investigating how the molecular diversity of hormones and their receptors arose (6, 7). One caution, regarding the phylogenetic positions of cyclostomes and chondrichthyans that are usually annotated as basal vertebrates, is that they have evolved from the common ancestor for the same amount of time as humans and thus should not necessarily have preserved the character state of the common ancestor (see 8, 9). They can thus exhibit some unique traits that were acquired uniquely in the cyclostome lineage. One intriguing example is the low osmoregulatory capacity of hagfish, which have a body fluid osmolality almost identical to that of seawater (10). This trait is shared with marine invertebrates and was regarded traditionally as the ancestral state of all extant vertebrates, but the widely accepted phylogenetic relationship of vertebrates (**Figure 2**) now suggests a secondary loss of this trait in the hagfish lineage after its ancestor diverged from the lamprey lineage.

3. EMERGING NON-OSTEICHTHYAN GENOME SEQUENCE INFORMATION

The presence and absence of orthologs must be judged carefully through phylogenetic approaches by using whole-genome sequences. Thanks to evolving massively parallel DNA sequence methods (11), the past 10 years have witnessed the release of whole-genome sequences of diverse species, including jawless and cartilaginous fishes (3, 12–18). Still, taxon sampling for these taxa is scarce and phylogenetically biased; thus, for cyclostomes, no genome-wide resources have been released

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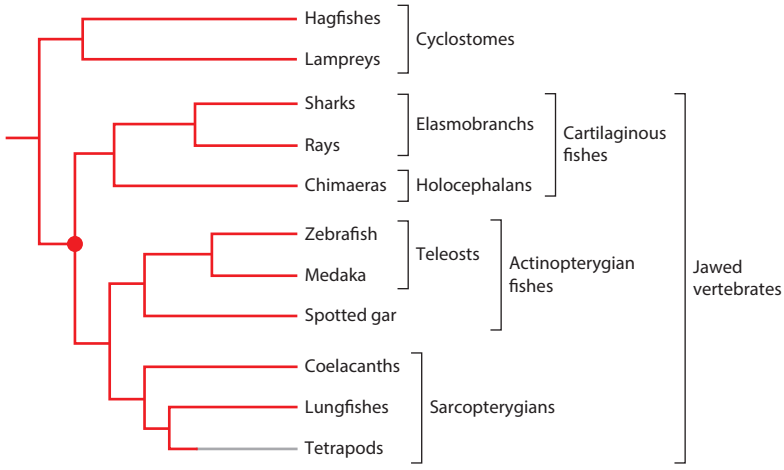

Figure 2

Phylogenetic relationships among different extant vertebrate lineages. The tree topology is based on existing literature supported by molecular phylogeny analysis (106). The circle indicates the Chondrichthyes–Osteichthyes divergence. Note that the only lineage including terrestrial species (*gray line*) is nested within the paraphyletic group of fishes.

for southern-hemisphere lampreys and hagfishes, as well as the entire superorder Squalomorphii among cartilaginous fishes.

As of May 2022, genome assemblies of five cyclostome species have been released in the National Center for Biotechnology Information (NCBI) Genome database (**https://www.ncbi.nlm.nih.gov/genome/**) after whole-genome sequencing (**Table 1**). Hagfish genome assemblies have been released for a single species, *Eptatretus burgeri*, as still-fragmentary but non-embargoed sequences from a ceased project (19) or as more complete sequences from a landmark project that remains unpublished at Ensembl (**https://ensembl.org/Eptatretus_burgeri/**; 20). Lamprey genomes have been sequenced and released for more species, namely the sea lamprey, *Petromyzon marinus* (12, 21); the Arctic lamprey, *Lethenteron camtschaticum* (14, 22); the Far Eastern brook lamprey, *Lethenteron reissneri* (13); and the Pacific lamprey, *Entosphenus tridentatus* (23), all annotated as chromosome-scale genome assemblies (**Table 1**). Although the effects of variable sizes and somatic rearrangement of the genomes are unclear (12), DNA sequencing and the assembly of resultant sequence reads suffer from abundant and lengthy repetitive elements, including those transferred supposedly from teleost fishes (e.g., 24), as well as highly variable GC-content of their sequences, as with lampreys (reviewed in 25).

Cartilaginous fish genome sequencing started with the elephant fish, *Callorhinchus milii*, classified in the subclass Holocephali even before 2010 (26), but it was not until 2017 that its substantial progress finally encompassed the subclass Elasmobranchii, the other taxon of the class Chondrichthyes (3, 27). As of May 2022, whole-genome sequences are available for more than five chondrichthyan species, some of which were produced by emerging consortium-type efforts listing chondrichthyan species for whole-genome sequencing [such as the Darwin Tree of Life (28), the DNA Zoo (29), Squalomix (3), and the Vertebrate Genomes Project (VGP) (30)]. Generally, chondrichthyan genomes exhibit large cross-species fluctuations in the interval lengths of intergenic and interexonic DNA sequences that complicate automated gene prediction (31).

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Table 1 Basic statistics of publicly available whole genome sequences of non-osteichthyan vertebrates[a]

Species	Number of sequences in the assembly	Total assembly length (Gb)	Proportion of sequences >1 Mb in length (%)	Proportion of remaining gaps (%)	N50 length (Mb)	# Seqs >1 Mb (haploid # of chromosomes[b])	Completeness of protein-coding regions[c] (%)	Reference
Cyclostomes								
Lethenteron reissneri	1,945	1.06	92.9	1.01	13.5	82 (N/A)	97.4	Zhu et al. (13)
Entosphenus tridentatus	141,124	0.93	74.2	13.95	10.8	74 (N/A)	87.6	Hess et al. (23)
Petromyzon marinus	1,434	1.09	87.5	1.38	13.0	88 (85)	97.9	Vertebrate Genomes Project (30)
Lethenteron camtschaticum	3,553	1.07	80.7	0.06	10.8	106 (85)	95.7	Nakatani et al. (22)
Eptatretus burgeri	10,846	2.61	84.9	44.97	2.7	835 (18)	63.9	Ensembl (20)
Chondrichthyans								
Amblyraja radiata	957	2.56	97.0	7.46	62.1	64 (50)	96.1	Rhie et al. (30)
Stegostoma tigrinum	1,266	2.77	99.5	6.20	76.6	50 (52)	94.4	Squalomix (100)
Scyliorhinus canicula	645	4.22	98.7	1.52	198.8	58 (32)	90.6	Darwin Tree of Life (28)
Callorhinchus milii	1,762	0.99	90.8	0.01	69.3	33 (N/A)	99.1	Nakatani et al. (22)

[a]See 52 for methodological concerns about completeness assessment.
[b]See 101 for details of the individual references for these karyotype reports.
[c]Completeness was measured with the proportion of the conserved single-copy orthologs identified as complete by the pipeline CEGMA (102) using the reference ortholog set CVG (103).

Typical metrics for evaluating genome sequence assembly quality show high completeness of protein-coding segments of the genomes, especially for lampreys (**Table 1**). Still, the genome sequences of some of these species show a significant gap between the total sequence length and the genome size estimated from the nuclear DNA contents measured by flow cytometry (31). Nevertheless, searching in these sequences—especially when involving multiple species—should guarantee high accuracy in the identification of gene repertoires in their genomes.

4. FREQUENTLY ENCOUNTERED DIFFICULTIES IN MOLECULAR PHYLOGENY INFERENCE

4.1. Tree Topology Discrepancies

One typical clue for interpreting the output of molecular phylogeny inference is to refer to widely accepted species phylogenies. Multiple occurrences of a particular subset in the inferred phylogenetic tree indicate gene duplication, whereas its absence indicates gene loss. Even more complicated is the discrepancy of tree topology even in the presence of a lineup of species assumed in the taxonomic group of interest. Such discrepancy can be explained by an altered rate of molecular sequence evolution in a particular taxon [referred to as long branch attraction (32)] or unusual sequence evolution beyond stepwise nucleotide substitutions such as gene conversion. The former difficulty can be overcome by applying molecular phylogeny methods that are robust against the heterogeneity of molecular evolutionary rates (reviewed in 33). Co-occurrence of an

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elevated molecular evolutionary rate and massive tandem gene duplications have been observed in the mouse locus encoding prolactin (PRL) that contains more than 20 copies duplicated uniquely in the rodent lineage (34, 35) (**Figure 1**). This gene family also underwent rather unexpected gene duplication and loss of the PRL gene in the tetrapod lineage, giving rise to the PRL1B gene retained uniquely in amphibians (36, 37). Gene conversion, which also makes the reconstruction of molecular phylogeny complicated, has been reported for some hormone-encoding genes, including the growth hormone (GH) gene and some members of the oxytocin (OT) and vasopressin (VP) gene family (38, 39).

4.2. Choice of Outgroup and Rooting

Another frequently encountered factor limiting our phylogenetic understanding of the evolutionary process of hormonal ligand molecules is the choice of outgroup sequences. Appropriate choice of outgroup facilitates reasonable rooting of the entire tree, which assists our understanding of how different subsets of the tree are interrelated. The difficulty of outgroup choice is encountered, for example, by phylogenetic studies for gene families with no explicit invertebrate orthologs available, including those of gonadotropin-releasing hormones (GnRHs) and kisspeptins (40), as well as the family containing prolactin (PRL or PRL1), PRL2, GH, and somatolactin (SL) (detailed below). Phylogeny inference of vertebrate hormone receptors also sometimes suffers from this problem, when only distantly related vertebrate paralogs are chosen as the outgroup. In this case, their immediate invertebrate orthologs, if available, should be used instead.

PRL modulates milk synthesis and mammary gland development in mammals (41). In diverse nonmammalian vertebrates, PRLs are reported to be involved in various physiological functions, such as regulating osmolarity, growth, and development (41). GH, which is homologous to PRL, stimulates the secretion of insulin-like growth factor-1 and promotes various metabolic processes, including bone and muscle growth (42). SL, another member of this family, is involved in the physiological control of body color, reproduction, and electrolyte balance in teleost fishes (43, 44). SL was long known only for teleost fishes, but its ortholog was later reported from coelacanths (3), suggesting that it is an ancient family member that arose before the osteichthyan–chondrichthyan split at the latest. PRL and GH are secreted in the anterior lobe of the mammalian pituitary gland, whereas SL is produced in the intermediate lobe (1). Some nonmammalian species also have retained an additional paralog of PRL encoding PRL2 (45). As of May 2022, only one cyclostome member of this gene family, namely the lamprey GH, has been reported (46) (NCBI GenBank ID, AB081461.1). Phylogenetic tree inference indicated the early expansion of this gene family before the chondrichthyan–osteichthyan split (3, 37, 47). Although this lamprey gene is designated as GH, it is not necessarily placed most proximally to the group of jawed vertebrate GH orthologs (3) and can also be regarded as orthologous to the SL group (**Figure 3*a***). The tree shown in **Figure 3*b*** assumes this lamprey gene, the only non-gnathostome member of the family, as the outgroup. A more recent report about another lamprey gene that belongs to this gene family demands further examination of molecular phylogeny (48). However, the absence of an explicit outgroup, ideally from invertebrates, prevents reliable rooting of the entire tree, so that the interrelationship between PRL, PRL2, GH, SL, and the only cyclostome family member identified so far remains unaddressed.

4.3. Homologs or Not?

Phylogeny inference with conventional methods of multiple sequence alignment assumes the common origin of the sequences included a priori. When attempting to infer the origin of, or the relationships among, individual gene families, sequences with much lower similarity must

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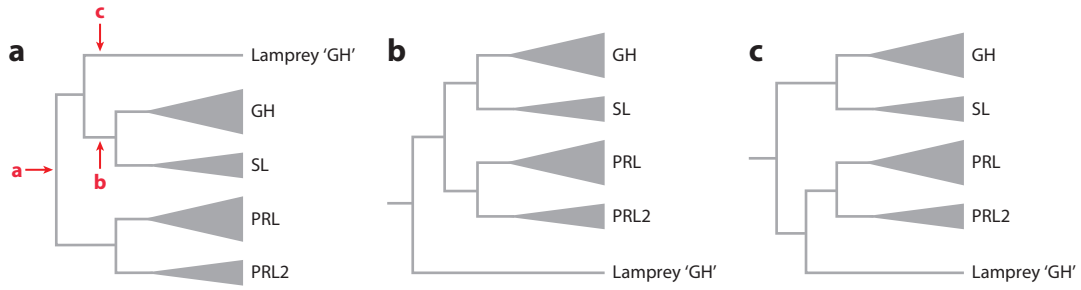

Figure 3

Indecisive rooting of the PRL/PRL2/GH/SL gene family. (*a*) Schematic presentation of the unrooted tree inferred and reported previously (3). Red arrows a, b, and c indicate potential positionings of the roots of the trees, which is reflected in panels *a*, *b*, and *c*, respectively. (*b*) Tree topology assuming the root position of the arrow b in panel *a*. (*c*) Tree topology assuming the root position of the arrow c in panel *a*. Abbreviations: GH, growth hormone; PRL, prolactin; SL, somatolactin.

be included in a data set. Traditionally, some of the hormones, including leptin and PRL, have been categorized into class-I helical cytokines, together with other signaling molecules, such as interleukin-6 and erythropoietin (49). These molecules are recognized as ligands by receptors containing variable numbers of fibronectin type-III domains, and the genes encoding those proteins are commonly categorized into the class-I cytokine receptor family (50). Similarities between those receptors are reasonably high for the genes encoding them to be recognized as homologs (genes with common descent). However, the similarity between ligand molecules in the class-I cytokines can be too low for their genes to be recognized as homologs. For example, the amino acid sequence of the human PRL cannot be aligned unambiguously with that of the human leptin, and the similarity at the amino acid sequence level is no more than 25%, with no strings of identical amino acids of more than two consecutive residues.

5. CREDIBILITY OF GENE ABSENCES

Even when genome sequences are released as whole genome information, they often harbor undetermined sequence regions and can even contain erroneously assembled sequence parts. For example, the publicly available reference genome sequences of the mouse GRCm39 (NCBI Genome ID, GCF_000001635.27) still have 286 stretches filled with N (if only those with a consecutive run of five Ns are counted) throughout the genome, and the proportion of Ns in its whole-genome assembly amounts to 2.7% of its total length. Even human whole-genome sequencing has been completed only for a selected cell line (51), and genome sequences being released for diverse species exhibit a wide spectrum of sequence completeness and quality. The completeness and quality of newly obtained genome sequences are usually assessed by dedicated tools for evaluation, and such sequences also undergo computation for simple statistics based on sequence length and compositions for quality control.

The credibility of any gene absence critically relies on the quality and completeness of the genome sequence resources that are consulted. When one attempts to evaluate the quality of whole-genome sequences, the coverage of conserved single-copy orthologs is often used as a metric that allows comparison of sequence quality between different species or different versions of the genomic sequence information. In fact, computational pipelines for assessing sequence completeness are not fully optimized for individual species. For example, in evaluating the sea lamprey genome sequences, some settings for assessment yield underestimated completeness scores, even when nucleotide or peptide sequences of predicted coding regions, which are regarded as refined

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Table 2 Variable assessment results of a whole-genome sequence assembly—a test case using the sea lamprey genome assembly[a]

Sequence type	Pipeline	Ortholog set	Completeness of protein-coding regions[b] (%)
Genome	BUSCO ver. 5	Vertebrata	62.4
		CVG	91.4
	CEGMA	CVG	97.9
Transcripts	BUSCO ver. 5	Vertebrata	77.7
		CVG	97.0
Peptides	BUSCO ver. 5	Vertebrata	79.2
		CVG	99.1

[a]The sea lamprey genome assembly kPetMar1.pri and its associated transcript and protein sequence data sets released at the NCBI Genome database (104) were used for evaluation.
[b]Completeness was measured with the proportion of the orthologs identified as complete, using computational pipelines BUSCO (105) and CEGMA (102).

after intron-aware gene prediction, are used as genome-wide input data (**Table 2**). Genome sequence assessment with this method often does not provide a full-mark completeness score even for the abovementioned "completely sequenced" human genome (52). These cases of completeness underestimation are attributable to the inherent difficulty of identifying protein-coding exons that are divergent in sequence because of large cross-species distances or intervention by long introns in vast genomic sequences. These difficulties can certainly be circumvented by careful manual identification of the sequence in question, while genome-wide completeness scores provide rough estimates of how reliable such ad hoc gene searches are with individual manual effort.

A famous example of elusive genes is the chicken ortholog of the leptin gene (53) (see below for details about this gene). The long-term quest for this gene was shown to have been foiled by the high GC content (54). Another example is the identification of prolactin receptor (PRLR) orthologs in elasmobranch genomes. The PRLR gene is a member of the class-I helical cytokine receptor family (50), and this gene, as well as the genes encoding the growth hormone receptor (GHR) and the somatolactin receptor, arose as a result of duplications in the vertebrate lineage (47). Previously, no chondrichthyan PRLR ortholog was reported (47), but our genomic survey identified partial sequences of a PRLR ortholog in three elasmobranch species (3). In addition, our transcriptome profiling identified the shark PRLR ortholog expression in the pituitary (3), as expected from some osteichthyan vertebrate species. The protein-coding nucleotide sequences of these shark genes have relatively high GC content of up to 70%, compared with the generally low and constant GC content of as low as 40% in the whole genomes of shark species (3). These examples show the difficulty of gene searches that is sometimes influenced by the genomic environment in which the gene is embedded, such as GC content, as well as transcription patterns, such as restricted spatiotemporal gene expression.

6. EXPANDED GENE REPERTOIRES REVEALED BY NON-OSTEICHTHYAN GENOMES

One interesting case of gene repertoire expansion unique to a subset of cartilaginous fishes was observed in the gene family containing VP and OT. VP and OT are homologous to each other and are peptide hormones produced in the hypothalamus and secreted from the posterior lobe of the mammalian pituitary (1). VP has physiological functions including water reabsorption in the kidney and blood pressure regulation by contraction of vascular smooth muscle cells (1, 55). OT regulates reproduction-related events, such as uterine contraction and stimulation of lactation,

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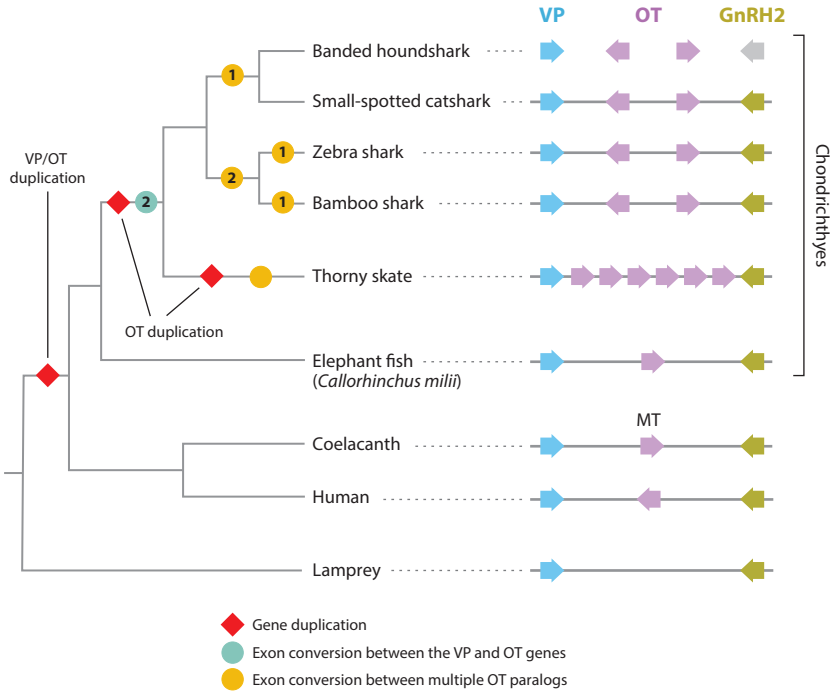

Figure 4

Modification to the vasopressin and oxytocin gene repertoires in the chondrichthyan lineage. The numbers in colored circles indicate which exon (1, 2, or 3) underwent gene conversion, whereas the open circle indicates gene conversion encompassing the entire coding region. The unidentified GnRH2 ortholog is indicated with a gray arrow. See Reference 3 for details. Abbreviations: GnRH, gonadotropin-releasing hormone; MT, mesotocin; OT, oxytocin orthologs; VP, vasopressin orthologs.

and is associated with social bonding (56). Naming of VP and OT peptides has conventionally been intricate, as it is based on amino acid sequences of their mature hormonal peptides encoded in the first exon (see 57), whereas the second exon encodes the carrier protein neurophysin (58).

The genes encoding VP and OT are proposed to have been generated in a tandem duplication of a single ancestral gene that probably existed in ancient jawless fishes and has also been retained in extant lamprey species (59) (**Figure 4**). This tandem duplication likely resulted in the conserved array of the orthologs of the VP and OT genes, as well as of GnRH2, in diverse vertebrates (1, 60) (**Figure 4**). For chondrichthyans, early studies revealed an expanded repertoire of VP and OT homologs in some shark species (61, 62). The first genomic investigation on elasmobranchs (3) and our scan of the genome sequences released later for more chondrichthyan species revealed putative open reading frames of one VP ortholog and two OT orthologs in the same genomic organization VP-OT-OT-GnRH2 (**Figure 4**).

An early study showed gene conversion between bovine VP and OT loci (e.g., 63). Inspired by this finding, molecular phylogeny was inferred separately for individual exons, which exhibited a remarkable discrepancy in overall tree topology between the obtained phylogenetic trees, suggesting repeated occurrences of exon-by-exon gene conversion (**Figure 4**). In the thorny skate (*Amblyraja radiata*) genome assembly released by the VGP (30), six tandem OT ortholog copies are located in a head-to-tail manner between the VP and GnRH2 orthologs (**Figure 4**), possibly as a result of tandem gene duplication unique to the lineages of batoids (rays and skates). These

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additional thorny skate OT copies have nearly identical nucleotide sequences in their coding regions, which suggests repeated gene conversions between them.

Overall, in the chondrichthyan lineages, gene conversions are implicated frequently between OT paralogs, and an ancient gene conversion between the VP and OT loci in early elasmobranch lineage was also suggested (**Figure 4**). Gene conversion of the whole exon 1, harboring the region encoding a mature hormonal peptide, is presumed to give rise to redundant hormone-producing loci with an identical peptide sequence encoded, but gene conversion of the exon 2 should not affect the hormonal peptide sequence. The physiological significance of such gene conversion (especially in the latter case) remains to be elucidated by future functional characterization of these genes.

As introduced above, elasmobranch genomes exhibit remarkably high genome size variation [haploid nuclear DNA content of 2.51 gigabase pairs (yellow guitarfish) to 16.7 gigabase pairs (angular roughshark); **https://www.genomesize.com/**; reviewed in 31]. Inherently, duplicated genes are frequently erased from genomes by constraints on genome size that can be costly in cellular processes including gene expression and DNA replication. More tandem gene copies observed in elasmobranchs indicate a possible trend of gene repertoire evolution in enlarged genomes in which duplicated genes can more frequently be retained, possibly to acquire a new function or spatiotemporal expression. This hypothesis that species with an enlarged genome can accommodate more tandem duplicates must be examined with more species with variable genome sizes.

7. NO SINGLE REPRESENTATIVE IN COMPARATIVE BIOLOGICAL STUDIES

The receptors for peptides encoded by the VP and OT homologs—VP/vasotocin receptor (VTR) and oxytocin receptor (OTR), respectively—are encoded by members of the seven-transmembrane G protein–coupled receptor gene superfamily. A revised nomenclature for VTR/OTR, as well as for their ligands VP/OT, has been proposed recently based on the VGP products; however, this took only the elephant fish (*C. milii*) into account in representing chondrichthyans (57, 64). By contrast, other studies encompassing elasmobranch species reported an additional gene repertoire whose orthologs are not retained by *C. milii*, namely VTR2C (or V2dR) and the abovementioned additional OT copy (3, 65). The two major extant groups of chondrichthyan lineages, namely the Holocephali and Elasmobranchii, sometimes exhibit explicit differences in retained gene repertoires from each other. These gene families illustrate the peril of representing a given taxon solely with its small subset of the entire diversity.

8. WHAT ROLE DID WHOLE-GENOME DUPLICATION PLAY?

High-continuity genome sequences allow not only comprehensive gene searches but also scrutiny of their genomic locations. Cross-species comparisons of gene locations assist our judgment of orthology by referring to the evolutionary history of neighboring genes, whereas paralogous comparisons within or across species highlight large-scale duplication between multiple arrays of genes. A plethora of studies on various gene families showed several genes that expanded into up to four copies in most cases, at an early phase of vertebrate evolution, which contributed largely to the long-established theory of two-round whole-genome duplication (WGD) (66–68). Some recent studies have adopted chromosome-scale genome sequences of non-osteichthyan vertebrates and invertebrates, which demonstrated the patterns of karyotype evolution; however, this resulted in rather inconsistent scenarios (22, 69, 70) (**Figure 5**).

Many genes encoding hormones and their receptors have been reported as part of such paralogons duplicated into multiple arrays of genes (e.g., 65, 71–73). Possibly one remarkable

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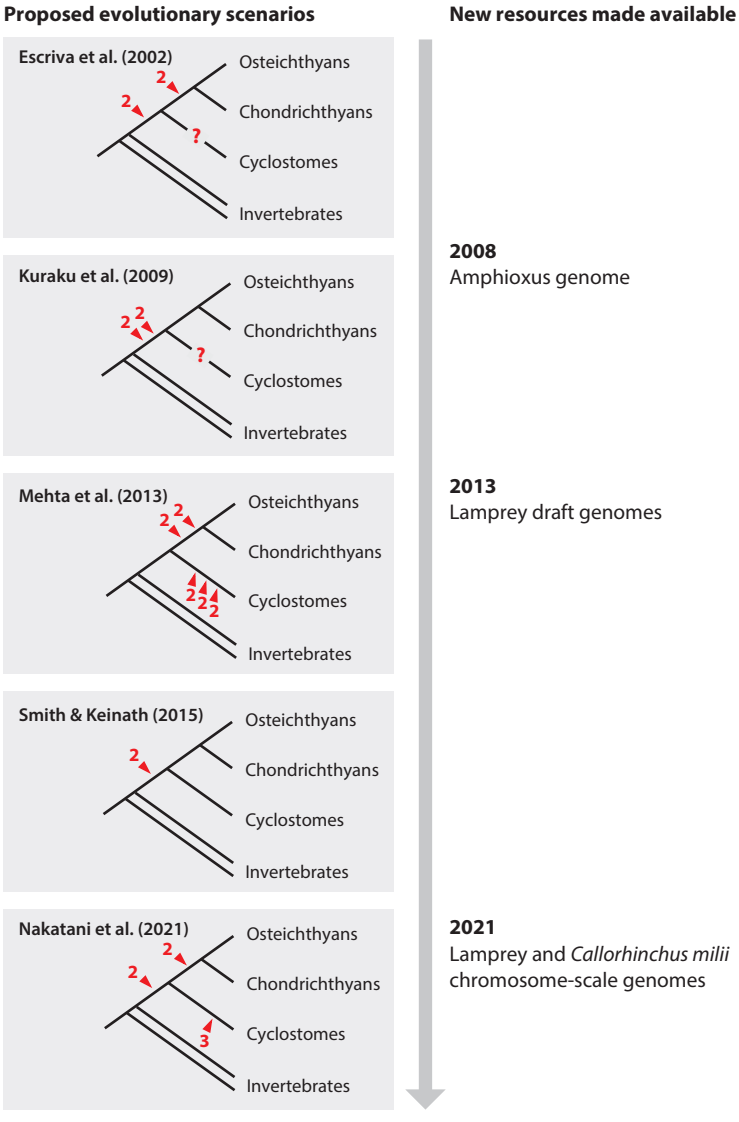

Figure 5

Equivocal scenarios of early vertebrate genome evolution. Different proposals of genome evolution based on phylogenetic analysis of vertebrate genes or genomes are depicted with the timeline shown along emerging genomic resources (14, 22, 107–110). Arrowheads on schematic trees indicate whole-genome duplications (WGDs). The occurrences of WGDs in the cyclostome lineage could not be inferred reliably before whole cyclostome genome sequences were made available, as indicated by question marks.

aspect of two-round WGD in the diversification of hormone-encoding genes is the synchronous multiplication of different hormone-encoding loci, which allows coevolution between multiple hormone-encoding genes or between a ligand and its receptor. One example is the genomic region containing genes encoding spexin, galanin, and kisspeptin that duplicated in early vertebrate evolution, followed by differential loss of paralogs from different duplicated copies (74). We must carefully examine whether gene duplications implicated in early vertebrate lineages are

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derived from WGD, whereas some of them can be explained by small-scale events, such as tandem gene duplication. One intriguing finding regards the aforementioned gene families encoding PRL/PRL2/GH/SL and their receptors. By dissecting the orders of whole-genome and small-scale gene duplications, ligand gene multiplications were shown to have occurred earlier than those of the corresponding genes encoding the receptors (47).

9. ASYMMETRIC EVOLUTION BETWEEN PARALOGS

As discussed so far, diversification of genes encoding hormones and their receptors exhibits a format that reflects the genome-wide pattern of molecular evolution. Still, vertebrate species in different evolutionary lineages have retained different gene repertoires. These repertoires are achieved in part through differential patterns of gene loss after the basic gene set was established, mainly through WGD. Again, this aspect of molecular evolution might also reflect genome-wide trends. Some recent studies have shown a genome-wide trend that genes are more frequently found to have been lost when they not only become functionally less important but also locate in genome regions susceptible to more mutations (75). The following section discusses a possible reflection of this genome-wide trend to the elusive nature of ghrelin and its receptor gene families.

Ghrelin is a 28-amino-acid-long gut hormone that regulates homeostasis, including appetite (76–78). This hormone is cleaved from a product of the preproghrelin gene (herein referred to as the ghrelin gene for simplicity), orthologs of which have been identified in diverse osteichthyans (**Figure 6**). Outside the taxon Osteichthyes, its potential orthologs, designated ghrelin-like genes, have been identified in some chondrichthyan species (3, 79, 80). Motilin is secreted from the upper intestine and is involved in gastrointestinal motility for digestion (81), and its orthologs have been identified in diverse gnathostomes, including elasmobranchs (3, 82) (**Figure 6**). Motilin transcripts of the shark species analyzed so far are localized in the intestine, as documented for mammals, whereas the shark ghrelin-like genes are intensively expressed in gastrointestinal tissues, including the stomach and pancreas, as well as in the central nervous system (3). Interestingly, cross-species synteny analysis suggested previously that the chondrichthyan ortholog of the conventional ghrelin gene was secondarily lost in its evolutionary lineage (3). Therefore, osteichthyan ghrelin and chondrichthyan ghrelin-like genes are not orthologous but paralogous to each other, as a result of reciprocal gene loss between the chondrichthyan and osteichthyan lineages (**Figure 6**), a phenomenon known as hidden paralogy (reviewed in 83).

The receptors for ghrelin and motilin are known as the GH secretagogue receptor (GHSR) and motilin receptor (MLNR), respectively, both of which belong to the large superfamily of seven-transmembrane G protein–coupled receptors. Previous genome survey and phylogenetic analysis identified elasmobranch orthologs for both GHSR and MLNR (3). This analysis revealed a quartet of paralogs in this receptor gene family, namely GHSR, MLNR, and two more paralogs, both of which underwent secondary loss in the lineage leading to mammals (3, 84, 85) (**Figure 6**). These two paralogs remain uncharacterized and are here tentatively designated as GHSR-like (for the ortholog group containing the coelacanth gene ENSLACG0000006806) and MLNR-like (for the ortholog group containing the green anole gene ENSACAG00000015346).

Both gene families (for ligand and receptor genes) harbor subtypes that exhibit frequent absence of orthologs as a result of gene loss in multiple independent lineages (**Figure 6**). For example, the motilin ortholog is absent from the mouse, rat, and western clawed frog genomes (86), possibly through lineage-specific loss, whereas their close relatives, namely squirrels (Sciuridae) in Rodentia and salamanders (Urodela) in Amphibia, have retained its orthologs (**Figure 6**). These gene loss–prone subtypes have a common feature of long branches in phylogenetic trees, suggesting elevated molecular evolutionary rates (3). Similar patterns have been observed in several gene

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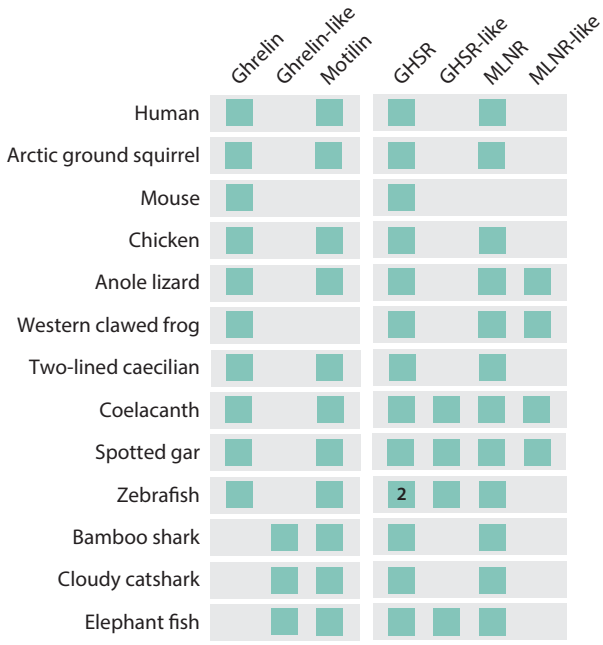

Figure 6

The elusive nature of the members of the gene families including ghrelin and motilin and the gene family including their receptors. Gene repertoires retained in individual genomes are shown with teal boxes based on the literature (3). The two zebrafish GHSR orthologs (and their teleost orthologs) are designated as GHSRa [also referred to as GHSR-LR (84, 85)] and GHSRb (111). Note that lampreys have retained one member for the ghrelin/motilin ligand gene family and up to three members of the GHSR/MLNR receptor gene family (3). Abbreviations: GHSR, growth hormone secretagogue receptor; MLNR, motilin receptor.

families harboring developmental regulatory genes (reviewed in 9): Genes with frequent loss during evolution are expressed in more spatiotemporally restricted manners, whereas others in the same gene family that are functionally indispensable and phylogenetically conserved are more widely expressed (75). Interestingly, gene fate asymmetry is reflected in not only gene expression patterns but also other aspects that are not directly related to the functions of individual genes. More concretely, from the genome evolution viewpoint, such elusive genes tend to be located in genomic regions with higher GC content and coverage of repetitive elements (75). Future investigations must examine whether these factors reflecting gene fate asymmetry also apply to genes encoding hormone ligands and receptors.

10. WHEN DID THAT GENE ORIGINATE? LEPTIN AS A POSSIBLE JAWED VERTEBRATE NOVELTY

The examples introduced so far depict typical patterns of gene repertoire evolution. Multiple paralogs in duplicated large-scale synteny blocks indicate that the gene and the array of the gene and its neighboring genes already existed before WGD.

In this sense, most of the hormone and hormone receptor genes analyzed in cartilaginous fish genomes were suggested to have existed in their vertebrate ancestor by the presence of putative cyclostome orthologs or gene duplication that probably occurred during WGD.

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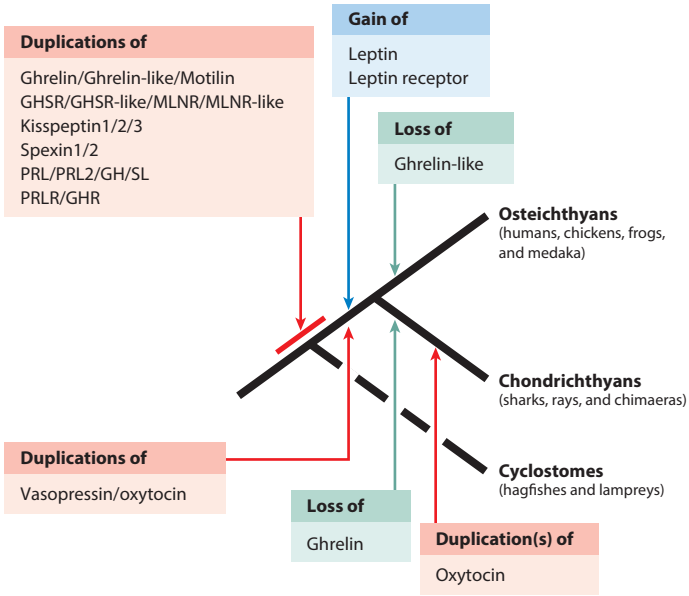

Figure 7

Deduced scenario of hormone and receptor gene repertoire evolution focusing on the earliest phase of vertebrate evolution. Phylogenetic timings of gene duplications, gene gains, and gene losses of hormone and receptor orthologs inferred in the literature (3) are indicated with the selected gene families introduced in the text. Abbreviations: GH, growth hormone; GHR, growth hormone receptor; GHSR, growth hormone segretagogue receptor; MLNR, motilin receptor; PRL, prolactin; PRLR, prolactin receptor; SL, somatolactin.

One remarkable exception is the gene encoding leptin, a key metabolic and neuroendocrine regulator in mammals (87), and the signaling cascade through its receptor (LepR) (88). Our previous survey identified orthologs of leptin and its receptor in both holocephalan and elasmobranch genomes through molecular phylogeny inference and conserved synteny, but no possible orthologs have been identified in cyclostome genomes (3). These observations—namely the absence of cyclostome orthologs and the absence of duplicated genes in any vertebrate—suggest that the signaling cascade involving leptin and its receptor LepR was an invention in the jawed vertebrate lineage (**Figure 7**). One remaining question is the ancestral role of this signaling cascade. In mammals, leptin is expressed almost exclusively in the adipose tissue, besides placenta (89). In zebrafish, leptin signaling is implicated in obesity control, but leptin expression is observed only in the central nervous system and digestive tract (90–92). Cartilaginous fishes generally lack overt adipose tissues (93), which has prevented identification of any tissue with intensive expression of its ortholog (3). In an in-depth sequence comparison of LepR, the amino acid residues responsible for downstream signaling modulation, such as S524 (serine at position 524 in the human LepR protein), Y986, Y1077, and Y1138, are shared in common by putative products of elasmobranch LepR orthologs (94–98). By contrast, of the residues known to participate in leptin binding (99), only L506 is conserved in elasmobranchs (3); however, this does not necessarily serve as negative evidence for ligand-receptor binding capacity, considering the low cross-species similarity of amino acid sequences of the leptin ligands (3). Although the function of the leptin signaling cascade in cartilaginous fishes remains uncharacterized, molecular evolutionary rates of the leptin ligand and its receptor slowed down particularly in the elasmobranch lineage (3), implying a high functional constraint on its roles in this unique lineage.

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SUMMARY POINTS

1. Emerging whole-genome sequence information is allowing a comprehensive survey of genes encoding hormones and their receptors.
2. Jawless and cartilaginous fishes are recognized as key taxa in understanding the diversification of hormone and receptor gene repertoires.
3. Surveys of whole-genome sequence information for multiple species in these taxa promise a high accuracy of gene repertoire identification in them.
4. The evolutionary patterns of the gene families including ghrelin and prolactin genes reflect the pattern of genome evolution.
5. The leptin gene, retained in chondrichthyan genomes and missing in cyclostome genomes, is possibly novel to jawed vertebrates.
6. Reduced constraint against genome size expansion in elasmobranchs might have allowed the retention of more tandemly duplicated genes, such as additional oxytocin genes.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

The authors thank Kazuaki Yamaguchi and Tingting Guo for their dedication to original data production, which led to the consolidation of viewpoints presented in this article. Our gratitude extends to John Donald, Wataru Takagi, and Yoko Yamaguchi for insightful discussions about the genes presented in this article; to Yoichiro Nakatani for suggestions in figure production; and to relevant researchers and technicians, including those working on the Vertebrate Genomes Project and Darwin Tree of Life for releasing unpublished genome sequences.

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